

Appendix 1 – Articles Description

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
1	IUCN greatly underestimates threat levels of endemic birds in the Western Ghats	(1) Model current extent of suitable habitat for 18 of the 26 Western Ghats endemic birds. (2) Quantify the disparity between ranges estimated by BirdLife International (BLI) and NatureServe for IUCN and ranges estimated using our SDM approach.	Boosted Regression Tree (BRT)	- Boosted Regression Tree (BRT) approach used to determine potential and presently available suitable habitat for each bird. - To assess the accuracy of the final model computed the area under the curve (AUC) and the Boyce index accuracy metric.	Western Ghats	18 endemic birds (Table 1) <i>Conservation status: 5 species listed as Endangered species, 4 Vulnerable species, 7 near threatened species and 2 least concern species</i>	- Habitat loss - Climate change	- eBird	2000-2015
2	Combined demographic and resource models quantify the effects of potential land-use change on the endangered Carnaby's cockatoo (<i>Calyptorhynchus latirostris</i>)	Calculate food resource availability for Carnaby's cockatoo using a daily ration model and integrate population viability analysis with alternative scenarios of potential resource change to evaluate the effects of foraging habitat loss on the population size and viability of Carnaby's cockatoo in the Perth–Peel region, while assuming no change in the extent or quality of breeding habitat, or in breeding and survival rates.	Daily ration model and integrate population viability analysis	- Daily ration model and integrate population viability analysis used to determine the carrying capacity. (Deterministic model of staged-based population growth to determine the annual mortality rates of juvenile and fledgling birds and Pentosidine assay method used to determine the adult survival rates. The relative importance of each life stage on population viability was determined by varying fecundity, immature or adult survival rates in the baseline model.) - A ceiling model of density dependence was used, whereby the population can grow only until it reaches the carrying capacity ceiling.	Perth–Peel region on the northern Swan Coastal Plain	Carnaby's cockatoo (<i>Calyptorhynchus latirostris</i>) <i>Conservation status: Endangered</i>	- Habitat loss (the loss of foraging resources through clearing of habitats for agriculture and urbanization)	Published literature, unpublished data and expert opinions (Table 1)	2001-2016

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3	Delineating and identifying long-term changes in the whooping crane (<i>Grus americana</i>) migration corridor	(1) Delineate a current migration corridor that varies geographically in size and shape and provides estimates of uncertainty, as any delineation is an estimate and requires some acknowledgment of uncertainty, using available whooping crane locations during migration. (2) Assess changes to the migration corridor that may have occurred over the past 8 decades.	General linear model	- GLM used to explain variation in opportunistic sightings over time. - Evoked a Bayesian analysis which used Markov chain Monte Carlo methods to explored temporal changes in characteristics of bird species. - Calculated means and variances from posterior probability distributions of each parameter and used these means and variances to define parameter priors as normal distributions in a subsequent and final analysis using telemetry locations.	Great Plains of North America	Aransas-Wood Buffalo Whooping cranes (<i>Grus americana</i>) <i>Conservation status: Endangered</i>	- Climate change - Land use change - Habitat loss (nesting) - Human influence (hunting)	Canadian Wildlife Service (CWS) and U.S. Fish and Wildlife Service (USFWS)	Opportunistic sighting data only collected from 1942-2016 and telemetry data collected from 2010-2016
4	Estimating conservation metrics from atlas data: the case of southern African endemic birds	(1) Examine issues of conservation interest, including the influence on birds of climate change. (2) Identification of non-climatic drivers of range change and changes in timing of migration as well as other questions of ecological interest.	Dynamic occupancy model	- Dynamic occupancy modelling used to test the population change metric against ‘probability of reporting’ change between time periods. - Examine correlation between population change scores and z-scores with probability of reporting change for each considered bird species at the QDGC level and then examine the relationship between the correlation coefficient output with the log-normalised number of QDGCs to examine the influence of range size on these comparisons.	South Africa, Lesotho and Swaziland	Considered only 58 from 69 endemics birds listed by Birdlife South Africa after excluding those with recent taxonomic splits (BLSA Checklist 2019) <i>Conservation status: Most of them are threatened or declining in population</i>	- Human influence - Climate change	The first and second Southern African Bird Atlas Projects - SABAP1 - SABAP2	1987-1992 2007-2017

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5	Examining the relationship between local extinction risk and position in range	Tested the hypothesis that local extinction rates are affected by distance to range edge and investigated whether effects are consistent among the Palaearctic and Indo-Malayan biogeographic realms, human-dominated versus unconverted land-use types, and species.	Multispecies Dynamic Bayesian occupancy-detection model	- The model is dynamic in that extinction and colonization of individual grid cells is modelled explicitly and multispecies in that fitted a single model to the full data set with species specific parameter estimates. - Used to assess the accuracy of the final model computed the relationship between extinction rates within cells (calculated as 1-persistence rates) and their position in range among realms and land-use type (unconverted and human dominated).	Palaearctic and Indo-Malay biogeographic realms	Galliformes 125 species <i>Conservation status: Over 25% of Galliformes are threatened (IUCN 2016)</i>	- Land-use change - Human influence (hunting)	Point-locality data collected from museums, journal articles, personal reports and letters, banding records, ornithological atlases Birdwatching trip-report websites	1727-2008
6	Habitat associations with counts of declining Western Grebes in Alberta, Canada	Examine the strength of the relationship between Western Grebe abundance (i.e., surveyed number of adults on a lake) and the relative probability of persistence on a suite of lakes that either historically supported or currently support grebes, and the relation between categories of Western Grebe abundance and habitat covariates (if persistence probability and abundance are not highly correlated, factors related to persistence might be different from those associated with abundance, thereby suggesting different implications for Western Grebe habitat conservation).	Logistic Regression	- Used to relate species abundance to habitat covariates. - Likelihood ratio test was used to compare each model to a null model without count predictors to test the significance of the model overall and alternative models were compared using AICc for small sample size, an information-theoretic approach.	43 lakes in Alberta, Canada where Western Grebes historically have occurred	Western Grebes (Aechmophorus occidentalis) <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Prey availability	Shoreline habitat surveys (Data on Western Grebe abundance were collected on 43 lakes at which the species was known to occur within the past 40 years in Alberta's Boreal Forest, Parkland, and Grassland regions)	2007-2009

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7	Migratory patterns and settlement areas revealed by remote sensing in an endangered intra-African migrant, the Black Harrier (<i>Circus maurus</i>)	(1) Describe for the first time the overall migration patterns (i.e. routes, distance, daily speed). (2) Identify the areas used during the breeding and non-breeding seasons. (3) Estimate the size of the settlement areas used by individuals during both the breeding and non-breeding seasons.	General Linear Mixed Models	- Used to evaluate whether the duration, distance travelled and average daily speed of migrants during a migration event differed according to sex or migration type (post-breeding vs. pre-breeding). - Response variables were fitted to a Gaussian distribution with an identity link function.	South Africa, Namibia and Lesotho	Black Harrier (<i>Circus maurus</i>) <i>Conservation status: Endangered</i>	Long migration distance	Location data collected from adult Black Harriers marked with GPS-GSM or PTT tracker devices during the breeding and non-breeding seasons	2008–2016
8	Combining data from multiple sources to design a raptor census - the first national survey of the Montagu's Harrier <i>Circus pygargus</i> in Poland	(1) Present the design, methodology and results of the first national quantitative survey of the Montagu's Harrier in Poland. (2) Show how data originating from various citizen science programmes can be used to design a rigorous survey at broad spatial scales. (3) Briefly discuss the implemented methods and provide recommendations for future surveys.	Predictive distribution model	- The random forest (RF) algorithm was used for building the predictive distribution model, represents a machine learning method based on classification and regression trees (CART). - The out-of-bag (OOB) samples are used to assess the prediction error of the procedure.	Eastern part of the country (Podlasie, Mazovia and the Lublin area) in Poland	Montagu's Harrier (<i>Circus pygargus</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	Human influence (agricultural intensification)	Presence-absence data obtained by volunteer-based citizen science projects conducted in Poland	2000-2012

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9	Non-invasive genetic sampling allows estimation of capercaillie numbers and population structure in the Bohemian Forest	(1) The current capercaillie population size. (2) The spatial and genetic structure and diversity of the capercaillie population living in the Bohemian Forest.	Bayesian clustering algorithms	- Bayesian clustering algorithms (using Markov Chain Monte Carlo (MCMC)) used to explore genetic population structuring and possible spatial genetic discontinuities. - Calculated F statistics and analysis of molecular variance (AMOVA) by assigning all genotypes to the population cluster according to the highest posterior probability. - Best model fit was selected using Akaike's information criterion (AIC).	Bohemian Forest (includes Šumava National Park in the Czech Republic and the Bavarian Forest National Park in Germany)	Western capercaillie (<i>Tetrao urogallus</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Human influence (increment of recreational activities)	Fresh capercaillie droppings on the entire potential capercaillie habitat in the Bohemian Forest	During 14 months from autumn 2009 to spring 2011
10	Citizen Science Reveals an Extensive Shift in the Winter Distribution of Migratory Western Grebes	(1) Continental patterns of population change. (2) Regional variation in population trends that might be indicative of a shift in the non-breeding distribution. (3) What changes in the oceanic environment might have led to regional patterns of population change?	Hierarchical model	Count data were modelled as hierarchical over-dispersed Poisson variables and models were fit using Markov Chain Monte Carlo (MCMC) methods.	Salish Sea of British Columbia and Washington State	- Western Grebes (<i>Aechmophorus occidentalis</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Clark's grebe (<i>Aechmophorus clarkii</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Climate change (Winter) - Habitat loss	- First analysis - Christmas Bird Count Data - Second analysis - Christmas Bird Count Data	1975–2010 1985-2010

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11	Annual variation in long-distance dispersal driven by breeding and nonbreeding season climatic conditions in a migratory bird	Two hypotheses were proposed to explain why later migrating individuals were more likely to disperse north of their natal site than individuals that migrated earlier. (1) Under the 'saturation' hypothesis, individuals select breeding sites based on the availability of suitable habitat. If the 'saturation' hypothesis is correct, predicted 1) that individuals that held territories in high quality non-breeding habitat the preceding winter would be more likely to disperse south of their natal grounds whereas individuals from low-quality territories would be more likely to disperse north. (2) Under this 'phenology' hypothesis, early migrating individuals will experience suitable breeding conditions and choose to settle at more southerly latitudes whereas later migrating individuals must continue migrating north to keep pace with the northward advance of resource phenology. If the 'phenology' hypothesis is correct, predicted 2) that northward dispersal of redstarts should increase in years when the onset of breeding-season phenology occurs relatively early and should decrease in years when breeding-season phenology occurs relatively late.	Mixed effect cumulative logit regression	- Used to quantify the effects of the predictors. - Used to test the significance of the fixed effects, removed each predictor from the full model and then tested the fit of the reduced model using a likelihood ratio test.	Patuxent Research Refuge in Laurel, Maryland	American redstarts <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Land use change (the phenology of breeding-season resource)	Stable hydrogen isotopes to quantify the rate and direction of long-distance immigration in a breeding population - Large-scale non-breeding habitat quality - mean normalized difference vegetation index from January to March from the major islands in the Greater Antilles (Jamaica, Cuba, Cayman Islands, Haiti, Dominican Republic, and Puerto Rico) - Breeding-season phenology - the peak blossom date of cherry trees on the National Mall in Washington - to quantify annual variation in the timing of phenology - Non-breeding territory quality - stable carbon isotope values from claw samples - to test prediction about the role of non-breeding territory quality	2009–2012

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12	Distinguishing local and global correlates of population change in migratory species	Distinguishing between local and remote correlates of change in seven migratory shorebird species while understanding how environmental conditions relate to population dynamics in migratory shorebirds of the East Asian–Australasian Flyway. Used citizen science data: firstly, to increase understanding of migratory population dynamics in declining populations of shorebirds; secondly, to illustrate how local citizen science monitoring data can be used to correlate local population dynamics with conditions elsewhere.	N-mixture model	- Zero-inflated N-mixture mode is used to estimate the abundance of all shorebird species (to account for variable observer effort and estimate yearly population growth rate). - Used weighted least squares (WLS) regressions to test for the effects of climatic anomalies at each migratory stage on local population growth rate. - Tested for species-specific responses in population growth rate to climatic anomalies at different migratory stages using AIC score.	Moreton Bay (Australia) and the East Asian–Australasian Flyway	- Bar - tailed godwit (<i>Limosa lapponica baueri</i>) <i>Conservation status: Near threatened</i> - Curlew sandpiper (<i>Calidris ferruginea</i>) <i>Conservation status: Least concern (population increasing)</i> - Eastern curlew (<i>Numenius madagascariensis</i>) <i>Conservation status: Endangered (Population decreasing)</i> - Great knot (<i>Calidris tenuirostris</i>) <i>Conservation status: Vulnerable (Population decreasing)</i> - Grey - tailed tattler (<i>Tringa brevipes</i>) <i>Conservation status: Near Threatened (Population decreasing)</i> - Lesser sand plover (<i>Charadrius mongolus</i>) <i>Conservation status: Least Concern</i> - Red knot (<i>Calidris canutus</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Climate change	Count data collected in seven migratory shorebird species from Moreton Bay	1992-2012
13	Factors influencing the distribution and abundance of endemic pigeons on São Tomé Island (Gulf of Guinea)	(1) The influence of environmental parameters on the distribution of pigeons. (2) Whether fruiting patterns of trees affect their distribution. (3) The phenology of distribution patterns. (4) Whether hunting pressure influences abundance.	General Linear Regression Models (GLM)	- Used to examine how canopy cover, number of large trees, habitat, season, indices of potential hunting pressure and fruit availability affected presence and abundance of each of the study species. - Used to test the null hypothesis that habitat structure was similar in different forest types, a nonparametric analysis of similarity (ANOSIM) was conducted on the raw environmental data. - Envfit function was performed to test for the significance of sampled variables and its relation to the ordination axes.	São Tomé Island (Gulf of Guinea)	- Four forest pigeons on São Tomé Island - Maroon Pigeon (<i>Columba Thomensis</i>) <i>Conservation status: Endangered (Population decreasing)</i> - São Tomé Green Pigeon (<i>Treron sanctithomae</i>) <i>Conservation status: Vulnerable (Population decreasing)</i> - Lemon Dove (<i>Columba simplex</i>) <i>Conservation status: Least Concern</i> - Bronze-naped Pigeon (<i>Columba malherbii</i>) <i>Conservation status: Near Threatened (Population decreasing)</i>	- Habitat loss - Human influence (hunting)	15 linear transects data - (total 35 km) in selected habitats and 10 shorter (1-km long) transects two trained observers during early mornings (between 05h00 and 11h00). Transects were walked at a slow pace (1 km/hour) and repeated twice in rainy and dry season respectively.	January and June 2010

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14	Fifteen years of annual Cape Parrot <i>Poicephalus robustus</i> censuses: current population trends and conservation contributions	Obtain an accurate population estimate of the Cape Parrot.	ANOVA	ANOVA used to analyse the CPBBD 15-year data set	Eastern Cape (including the former Transkei), Limpopo, Mpumalanga and KZN	Cape Parrot (<i>Poicephalus Robustus</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Human influence (illegal pet trade)	- Presence at forests and at feeding sites outside of forests were collected opportunistically from observers including landowners, birders and conservation managers. - CPBBD 15-year data set	1998-2012
15	Isolation and decline of a population of the Orange-Breasted Falcon	(1) Present demographic data recorded during two decades of study in Belize and Guatemala. (2) Summarize what is known about the species' past and present distribution in Central and South America, to examine geographic and temporal trends across its entire range.	General Linear Mixed Models (GLMMs)	- Models tested for changes in occupancy and the number of fledglings. - Ecological niche modelling used to examine trends in the species' distribution, generated predictive habitat maps from bioclimatic, topographic, and land-use variables derived from distribution records. - Baker model used to simulate the effects of changes in mortality, fecundity, and carrying capacity, to estimate its probability of extinction.	Mountain Pine Ridge in western Belize	Orange-Breasted Falcon (<i>Falco deiroleucus</i>) <i>Conservation status: Near Threatened (Population decreasing)</i>	- Human influence (human related conflicts) - Habitat loss (competition for nest sites) - Predation (depredation by increasing numbers of Black Vultures (<i>Coragyps atratus</i>))	- Demographic data of the eyries Baker studied (1992–1997) to the same territories in Belize (2003–2009) - All known and monitored territories in the Petén of Guatemala (2008–2009)	1992-2009
16	Habitats and landscapes associated with bird species in a lowland conifer-dominated ecosystem	Examined the characteristics of these species' breeding habitat by developing and testing a method for modelling habitat suitability across the lowland conifer forests of the Agassiz Lowlands Ecological Subsection (ALS).	Predictive models	- Used MaxEnt to develop predictive models and maps of boreal bird distributions and to model correlations between specific species' presence locations and environmental predictor variables. - Statistical significance was determined using chi-square tests.	The lowland conifer forests of the Agassiz Lowlands Ecological Subsection in northern Minnesota	- Spruce Grouse (<i>Falcipennis canadensis</i>) - <i>Conservation status: Least concern (Population stable)</i> - Black-backed Woodpecker (<i>Picoides arcticus</i>) - <i>Conservation status: Least concern (Population stable)</i> - Olivesided Flycatcher (<i>Contopus cooperi</i>) - <i>Conservation status: Near threatened (Population decreasing)</i> - Yellow-bellied Flycatcher (<i>Empidonax flaviventris</i>)	Climate change	- Point counts conducted in 130 forest stands of the ALS (The Agassiz Lowlands Bird Project (ALBP)) and opportunistic observations gathered during	2014

				- MaxEnt models were generated with each variable and evaluated using the area under the receiver operating curve (area under curve; AUC) as a test of the variables' capacity to separate species occurrence locations from random background locations. - The model with the lowest AICc value was selected as the best model and was used for interpretation and mapping.		<i>Conservation status: Least Concern (Population increasing)</i> - Boreal Chickadee (<i>Poecile hudsonicus</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Golden-crowned Kinglet (<i>Regulus satrapa</i>) <i>Conservation status: Least Concern (Population increasing)</i> - Ruby-crowned Kinglet (<i>Regulus calendula</i>) <i>Conservation status: Least Concern (Population increasing)</i> - Swainson's Thrush (<i>Catharus ustulatus</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Connecticut Warbler (<i>Oporornis agilis</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Palm Warbler (<i>Setophaga palmarum</i>) <i>Conservation status: Least Concern (Population increasing)</i> - Dark-eyed Junco (<i>Junco hyemalis</i>) <i>Conservation status: Least Concern</i>		field data collection - The Minnesota Breeding Bird Atlas project (MNBBA; http://mnmba.org/)	
17	Life history traits and niche instability impact accuracy and temporal transferability for historically calibrated distribution models of north American birds	Examine the relationship between life history characteristics and model accuracy and transferability for a subset of North American birds (use historical occurrence records in combination with historical climate data to forecast to the current climate and to validate the models).	Environmental Niche Models	- The relationship between life history characteristics and Environmental Niche Models (ENMs) was assessed using GLMs. - Interpolative models (Im), are developed using occurrence data from a particular decade in conjunction with historical climate data from the same decade and are expected to be good representations of the species' distribution. - Extrapolative models (Em), which are the interpolative models projected onto a 'future' climate are more prone to error because they are built using climate data that differs from the climate to which they are projected. - MaxEnt used to build models and then project them into the "future". - Built models in order to test the temporal transferability of ENMs. - Validated the models using AUC which is an appropriate performance measure.	North America	Species studied: Table 1 <i>Conservation status: Majority are threatened or declining in population</i>	Climate change Habitat loss	- Historical occurrence and climate data (1950-1990s) to build models for a sample of birds, and then projected them forward to the 'future' (1960-1990s) - Frequency maps using eBird - Survey data from the National Audubon Society's Christmas Bird Count (CBC)	1950-1999

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18	Incorporating abundance information and guiding variable selection for climate-based ensemble forecasting of species' distributional shifts	(1) Test whether performance of ENMs improved when using only variables known to directly affect species' abundance, compared to performance of models based on other variable selection approaches. (2) Use the top performing ENMs from our variable selection exercise to analyse species abundance data in relation to future distribution shift estimates to identify potential critical areas of loss in environmental suitability.	Maximum entropy modelling	- Used maximum entropy algorithm Maxent, to test which variable selection approach performed best for creating Ecological niche models (ENMs). The Maxent algorithm is used for generating ENMs with presence-only/pseudoabsence data and climatic variables. - Used the average Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) to evaluate the accuracy of the models.	North America	North American quails - California quail (<i>Callipepla californica</i>) <i>Conservation status: Least Concern</i> <i>(Population increasing)</i> - Gambel's quail (<i>Callipepla gambelii</i>) <i>Conservation status: Least Concern</i> - Scaled quail (<i>Callipepla squamata</i>) <i>Conservation status: Least Concern</i> <i>(Population decreasing)</i> - Northern bobwhite (<i>Colinus virginianus</i>) <i>Conservation status: Near Threatened</i> <i>(Population decreasing)</i> - Montezuma quail (<i>Cyrtonyx montezumae</i>) <i>Conservation status: Least Concern</i> <i>(Population stable)</i> - Mountain quail (<i>Oreortyx pictus</i>) <i>Conservation status: Least concern</i>	Climate change	Species occurrence data from the BBS and eBird databases	1966–2000 and 1950–2000 respectively

19	Opposing responses to drought shape spatial population dynamics of declining grassland birds	Examined long-term spatial trends and asked whether regional abundance was influenced by drought conditions at two temporal scales, the month of arrival on the breeding grounds and over the preceding 4 years and tested whether environmental conditions experienced along the migration pathway influenced range dynamics. Show evidence for spatially autocorrelated abundance across the range of a latitudinal migrant due to the influence of environmental conditions along the migration pathway on breeding settlement patterns.	Hierarchical Bayesian Model	- The count data were modelled as hierarchical over dispersed Poisson variables and models were fit using Markov chain Monte Carlo methods.	Great Plains, North America	- Lark Bunting (<i>Calamospiza Melanocorys</i>) <i>Conservation status: Least concern</i> - Chestnut-collared Longspur (<i>Calcarius ornatus</i>) <i>Conservation status: Near Threatened (Population decreasing)</i>	- Climate change - Land-use change	48 years of BBS data which is conducted annually during the breeding period in late May to early July.	1967–2014
20	Persistence of the threatened Knysna warbler <i>Bradypterus sylvaticus</i> in an urban landscape: do gardens substitute for fire?	Compare prey availability between warbler-occupied and unoccupied habitats.	ANOVA	- For habitat comparisons, repeated-measure analysis of variances (ANOVA) was calculated on the bootstrap means, with post-hoc uncorrected pair-wise comparisons. Repeated-measure analyses were used to reduce the effect of season on site data. - Seasonal variation was analysed using pair-wise Kruskal–Wallis nonparametric ANOVAs.	Table Mountain – Table Mountain National Park (TMNP), Cape Peninsula, South Africa	Knysna warbler (<i>Bradypterus Sylvaticus</i>) <i>Conservation status: Vulnerable</i>	- Habitat loss - Prey availability	Representative cross-section of invertebrates, pitfall trapping, quadrat searches and Berlese-Tullgren funnel litter extractions were used.	Sampling was carried out four times from July 2005 to April 2006
21	Population size and trends of Ludwig's Bustard <i>Neotis ludwigii</i> and other large terrestrial birds in the Karoo, South Africa	Repeat Allan's (1994) road and aerial counts to assess population trends for Ludwig's Bustard and other large terrestrial birds of the Karoo over the past two decades, and to seek evidence to support any of his theories regarding the apparently constant Nama Karoo population.	Generalised Linear Models and Generalised Additive Models	- Assessed the effect of rainfall in the preceding months on the numbers of birds per 100km seen on road counts with Pearson's product-moment correlation coefficients, Generalised Linear Models and Generalised Additive Models. - Mann-Whitney and U-tests were used to test the effect of season on average group sizes. - Wilcoxon signed-rank tests used to investigate the effect of road surface on the numbers of bird species seen.	Karoo, South Africa	- Ludwig's Bustard (<i>Neotis ludwigii</i>) <i>Conservation status: Endangered (Population decreasing)</i> - Large terrestrial birds	Human influence (high collision mortality on power lines)	- Extensive road and aerial census counts - Allan's (1994) original counts - the 1980s counts (available from DGA) - Road counts followed two transects in the western and eastern Karoo four times over one year with two aerial counts - the 2010s counts	Late 1980s (1989) 2010-2011 (2012)

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22	Projected impacts of climate change on protected birds and nature reserves in China	(1) Explore the potential impact of climate change. (2) Quantify and map the spatial pattern of species range shift. (3) Evaluate the potential effectiveness of nature reserves in China under climate change.	Maximum entropy model	- Adopted Maxent based on the principle of maximum entropy, to predict potential distribution since Maxent is one of the best presence-only Species distribution modelling (SDMs) which is robust to biased samples. - Used the average Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) to evaluate the accuracy of the models.	Mainland of China including Hong Kong (China for short); all islands were removed except for Hainan Island	108 breeding species of Chinese protected birds including 56 species of threatened birds and 11 endemic birds	- Climate change - Habitat loss	China Bird Report (CBR, http://birdtalker.net/)	2003-2007
23	Population size, breeding performance and habitat use of the Black-winged Pratincole <i>Glareola nordmanni</i>	Following main factors in the breeding areas: (1) Ploughing of steppe (more than 25 million ha in Kazakhstan) resulting in habitat loss and low breeding success in colonies on arable fields. (2) Trampling of clutches and chicks due to lower mobility and stronger concentration of livestock compared to previous times. (3) Higher predation rates by increasing numbers of corvids. (4) Climate change with increasing frequencies of extreme weather (e.g. hailstorms) and droughts.	Binary logistic regression	- Used to analyse habitat selection. - To assess the predictive power of the final models, several measures of model quality were calculated: i) Nagelkerke's R², which describes model calibration and refinement, ii) AUC (area under the receiver operating characteristic, ROC), which describes discrimination and iii) Cohen's Kappa, which describes discriminative power prevalence-independent.	Lake Tengiz, 120 km SW of Astana, Kazakhstan	Black-winged Pratincole (<i>Glareola nordmanni</i>) <i>Conservation status: Near Threatened (Population decreasing)</i>	- Habitat loss (trampling of clutches and chicks) - Predation - Climate change	- Korgalzhyn study area was surveyed in breeding season. - The Lower Irtysh study area was surveyed. - Surveyed all previously known breeding sites of Black-winged Pratincole, on which information was available from local databases and expert communication. - All wetland sites similar to habitat used in previous years (freshwater lakes with open shores, solonchaks). - Large areas of fallow fields and pristine steppe (done from a vehicle).	End of April to the end of July 2006 End of April to the end of July 2007

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
24	Quantifying the decline of the Martial Eagle <i>Polemaetus bellicosus</i> in South Africa	Quantifying the decline of the Martial Eagle <i>Polemaetus bellicosus</i> in South Africa	Generalized linear models	- Used to quantify changes in reporting rates between the two SABAP projects. - The significance (Chi-square test statistic and P-values) of the explanatory terms in the models were explored using the 'ANOVA' function applied to the output from the GLMs.	South Africa	Martial Eagle (<i>Polemaetus bellicosus</i>) <i>Conservation status: Vulnerable</i>	- Human influence (population growth in Africa) - Habitat loss - Land use change	Data on Martial Eagle occurrence from first and second Southern African Bird Atlas Projects; - SABAP 1 - SABAP 2	1987–1992 2007–2012
25	Rapid population decline in migratory shorebirds relying on Yellow Sea tidal mudflats as stopover sites	Measure abundance and population trends for ten EAAF shorebird taxa for which expert-defined migratory connectivity networks were available to test if taxa that rely heavily on Yellow Sea tidal mudflats to stage their long-distance migrations are experiencing the most rapid population declines.	N-mixture model	- N-mixture model formulated as a hierarchical Poisson regression that estimates parameters for the latent state of abundance by accounting for imperfect detection of individuals in spatially or temporally replicated counts. - The proportion of Markov Chain Monte Carlo (MCMC) samples where discrepancies in the observed data exceed those of the replicate data should be near 0.50 when model fit is adequate.	Yellow Sea tidal mudflats - Australia and New Zealand at 43 sites organized into four nodes of the migratory connectivity network	Most abundant migratory shorebirds: - Bar-tailed godwit (two subspecies here analysed separately: <i>Limosa lapponica baueri</i> and <i>Limosa lapponica menzbieri</i>) <i>Conservation status: Least Concern</i> <i>(Population decreasing)</i> - Far eastern curlew (<i>Numenius madagascariensis</i>) <i>Conservation status: Endangered</i> <i>(Population decreasing)</i> - Terek sandpiper (<i>Xenus cinereus</i>) <i>Conservation status: Least Concern</i> <i>(Population stable)</i> - Grey-tailed tattler (<i>Tringa brevipes</i>) <i>Conservation status: Near Threatened</i> <i>(Population decreasing)</i> - Great knot (<i>Calidris tenuirostris</i>) <i>Conservation status: Vulnerable</i> <i>(Population decreasing)</i> - Red knot (two subspecies here analysed together:	- Human influence (human-induced global change) - Long migration distance	Non-breeding season counts	1993-2012

						Calidris canutus piersmai and Calidris canutus rogersi) <i>Conservation status: Least Concern</i> <i>(Population decreasing)</i> - Curlew sandpiper (Calidris ferruginea) <i>Conservation status: Least Concern</i> <i>(Population increasing)</i> - Lesser sand plover (Charadrius mongolus) <i>Conservation status: Least Concern</i>			
26	Regional patterns of habitat use by a threatened forest bird, the Bicknell's Thrush (<i>Catharus bicknelli</i>), in Quebec	(1) Analyse geographic variation in habitat use by Bicknell's Thrush, based on multiyear surveys in two regions of its Canadian range. (2) Assessed the relative roles of habitat, topography, tree composition, silvicultural practices affecting the structure of the habitat, ground cover, and elevation and its variation at local and two spatial scales, as well as Swainson's Thrush abundance, on abundance of Bicknell's Thrush. (3) Tested whether Bicknell' Thrush abundance is higher (i) where precommercial thinning has not occurred, (ii) at higher elevations or in steeper terrain, and (iii) when Swainson's Thrush abundance is relatively low. (4) Discuss how the results of this study may inform future efforts to designate and manage this threatened species' critical habitat over its entire breeding range.	Generalized linear models	- Used generalized linear models (GLMs) with a negative binomial error structure with point-count stations as repeated-measure subjects. - Quasi-likelihood information criterion corrected for small sample size (QAICc) used for the Mont Gosford multiyear data set and Akaike' s information criterion corrected for small sample size (AICc) used for the Massif du Sud single-year data set.	Both in the Appalachian Mountains range: Mont Gosford Quebec and Massif du Sud	Bicknell's Thrush (<i>Catharus bicknelli</i>) <i>Conservation status: Vulnerable (Population decreasing)</i>	Habitat loss	- Mont Gosford multiyear data set - Bicknell's and Swainson's thrushes at; 99 point-count stations 129 stations - Point counts	1980s and up to 2007 2001, 2003, 2004, and 2007 2007

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
27	Significance of the globally threatened Straw-headed Bulbul <i>Pycnonotus zeylanicus</i> populations in Singapore: a last straw for the species?	(1) Reviewed the status of the Straw-headed Bulbul across Singapore by synthesising historical and current information from varied sources. (2) Estimated the population trends of the species based on our long-term data. (3) Estimated the present population of the species in Singapore (Singapore and Ubin islands).	Log-linear regression models	- Log-linear regression models used to model the population trends for the data, used log-linear regression models with a Poisson-error distribution. - Used to estimate the goodness-of-fit for each model, the chi-square statistic and likelihood ratio for each model, and its associated P value were computed.	Singapore and Ubin islands	Straw-headed Bulbul (<i>Pycnonotus zeylanicus</i>) <i>Conservation status: Vulnerable (Population decreasing)</i>	Human influence (regional bird trade)	Count data in our study was compiled from two major citizen science initiatives; - The Annual Bird Census (ABC) - The Mid-year Bird Census (MBC)	for 30 years (since 1986) for 15 years (since 2000)
28	Species-dependent effects of habitat degradation in relation to seasonal distribution of migratory waterfowl in the East Asian–Australasian Flyway	Quantified the spatial patterns of habitat degradation in wetland sites, in relation to the seasonal distributions of eight waterfowl species.	General linear model	- Three General linear models (GLMs) were applied to test (1) whether patterns of wetland degradation change over latitude and (2) whether the patterns differ among species with different migration extent (i.e., species with shorter and broader migration corridors versus those with longer and narrower migration corridors). - Best logistic regression models were chosen using AIC values.	East Asian–Australasian Flyway	- Tundra swan (<i>Cygnus columbianus</i>) <i>Conservation status: Least concern</i> - Swan goose (<i>Anser cygnoides</i>) <i>Conservation status: Vulnerable (Population decreasing)</i> - Bean goose (<i>Anser fabalis</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Greater white-fronted goose (<i>Anser albifrons</i>) <i>Conservation status: Least Concern</i> - Lesser white-fronted goose (<i>Anser erythropus</i>) <i>Conservation status: Vulnerable (Population decreasing)</i>	Habitat loss	- eBird - Land cover maps for from the European Space Agency (ESA) CCI 300-m annual global land cover products (European Space Agency 2017) to quantify the spatial patterns of habitat degradation.	1992-2016 1992 and 2012

						- Greylag goose (<i>Anser anser</i>) <i>Conservation status: Least Concern</i> (<i>Population increasing</i>) - Falcated duck (<i>Anas falcata</i>) <i>Conservation status: Near Threatened</i> (<i>Population decreasing</i>) - Baikal teal (<i>Sibirionetta formosa</i>) <i>Conservation status: Least concern</i> - Spot-billed duck (<i>Anas poecilorhyncha</i>) <i>Conservation status: Least Concern</i> (<i>Population decreasing</i>) - Northern pintail (<i>Anas acuta</i>) <i>Conservation status: Least Concern</i> (<i>Population decreasing</i>)			
29	A hierarchical spatial model of avian abundance with application to cerulean warblers	Current methods poorly accommodate both observer and spatial effects because modelling these spatially autocorrelated counts within a hierarchical framework is not practical using standard statistical approaches. Therefore, in this study, propose a Bayesian approach to this problem and provide as an example of its implementation a spatial model of predicted abundance for the Cerulean Warbler (<i>Dendroica cerulea</i>) in the Prairie-Hardwood Transition of the upper midwestern United States.	Hierarchical Model	- Present a hierarchical spatial model which is a spatial version of the objective Bayesian hierarchical model with environmental covariates for regional prediction of abundance data and fit the model to data with Markov chain Monte Carlo methods (MCMC).	Prairie-Hardwood Transition of the upper midwestern United States	Cerulean Warbler (<i>Dendroica cerulea</i>) <i>Conservation status: Vulnerable (Population decreasing)</i>	- Land-use change (land cover composition and configuration and terrain heterogeneity) - Climate change - Human influence	21 years of North American Breeding Bird Survey counts	1981-2001

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
30	Using citizen-science data to identify local hotspots of seabird occurrence	The primary objective of the analysis was to apply spatiotemporal models to data from the Puget Sound Seabird Survey to; (1) Evaluate relative hotspots of occurrence over the period 2007–2014. (2) Evaluate temporal trends in occupancy over this period. (3) Establish a baseline for future monitoring in the region.	Spatial occupancy model	- Constructed a spatial occupancy model separately for each species to incorporate spatial patchiness as well as annual and seasonal variation fit the model to data with Markov chain Monte Carlo methods (MCMC).	Puget Sound region (Washington State, USA)	18 seabird species including; - Marbled murrelet (<i>Brachyramphus marmoratus</i>) <i>Conservation status: Endangered (Population decreasing)</i> - Brandt's cormorant (<i>Phalacrocorax penicillatus</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Pelagic cormorant (<i>Phalacrocorax pelagicus</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Surf scoter (<i>Melanitta perspicillata</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Red-throated loon (<i>Gavia stellate</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Common loon (<i>Gavia immer</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Western Grebe (<i>Aechmophorus occidentalis</i>) <i>Conservation status: Least Concern (Population decreasing)</i> - Horned grebe (<i>Podiceps auratus</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Climate change - Human influence	The Puget Sound Seabird Survey of the Seattle Audubon Society, is unique in that it represents a seven-year study, collected with a focus on winter months (October–April)	2007-2014

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
31	When and where to move: Dynamic occupancy models explain the range dynamics of a food nomadic bird under climate and land cover change	The goal of the study was to investigate 13 years of colonization and extinction dynamics of Cape parrot as a function of resource distribution and changes in climate and land cover along its range.	Dynamic occupancy model	- Applied dynamic occupancy model to evaluate how the aforementioned covariates were linked to colonization of unoccupied sites and extinction of occupied sites by bird species. - Best model fit was selected using Akaike's information criterion (AIC).	South Africa	Cape parrot (Poicephalus robustus) <i>Conservation status: Least Concern (Population decreasing)</i> A food nomadic and endemic bird	- Land use change - Climate change	Cape Parrot Big Birding Day (CPBBD) data - citizen science driven presence/absence data	2002-2014
32	Will the California Current lose its nesting Tufted Puffins?	Use this colony occupancy data to model the relationship between nesting habitat and current environmental conditions to project future suitable breeding sites in the same geographic range. Present these results as an example of how this information can be used in both regulatory (e.g., Endangered Species Act) and conservation planning contexts.	Hindcast models (GLM, GBM and RF)	- Hindcast models were created to examine the past relationship between temperature and patterns of bird species colony occupancy. Hindcasts can increase confidence in future projections and help shed light on ecological interactions over time. - To help acknowledge and estimate uncertainty, three different models using different statistical approaches were selected; generalized linear models (GLM), generalized boosting models (GBM, also referred to as boosted regression trees) and random forests (RF). The GLM models used a logit link between the response variable mean and combination of explanatory variables (i.e., logistic regression). GBMs incorporate regression and machine-learning techniques through boosting many decision-tree models to increase model performance. Decision models recursively partition sets of explanatory and outcome variables in a stagewise manner until subsets of data are explained by trees of bifurcating decisions. The differences in	California, USA	Tufted Puffin (Fratrula cirrhata) <i>Conservation status: Least Concern (Population decreasing)</i>	- Human influence (large marine ecosystem) - Climate change - Habitat loss	- WorldClim - a set of global climate layers derived from interpolation of monthly climate observations. - Radiative Concentration Pathways (RCP) 4.5 and 8.5 as defined by the IPCC 5th Assessment Report as future environmental projections against which to forecast changes in Tufted Puffin breeding distribution. - Species distribution data - Washington Department of Fish and Wildlife and Environment and Climate Change Canada and were derived from	1950-2000 All for the year 2050 (average of 2041-2060) 1950-2009

				statistical and machine learning approaches across GLM, GBM and RF algorithms provides variance across which to test sensitivity between models as well as estimations of model uncertainty. - The area under the receiver operating characteristic curve (AUC) and the True Skill Statistic (TSS) were the two-model metrics used to evaluate model performance.				expansive US and Canada breeding colony surveys conducted by groups including USFWS, Washington Department of Fish and Wildlife and others.	
33	Using Latent Selection to Model Persistence in a Declining Population	Hypothesize that factors known to be important to the western grebe, such as ample nesting habitat, prey availability, and adequate lake size, would be positively correlated with western grebe persistence, while factors that might cause disturbance to the birds during their breeding season (e.g. human activity and development) would be inversely correlated with persistence.	Persistence models	- Alternative biologically plausible models were assessed using AICc for small sample size.	Alberta	Western grebe (<i>Aechmophorus occidentalis</i>) <i>Conservation status: Least Concern (Population decreasing)</i>	- Habitat loss - Human influence - Land-use change - Prey availability	Survey data - used 43 publically accessible lakes for the presence of western grebes in the Boreal forest up to three times (roughly once every 30 days) and documented human developments, shoreline/emergent/backshore vegetation and other lake characteristics perceived to be important for the western grebe.	From late May through late August of 2008 and 2009
34	Assessing predictions of population viability analysis: Peregrine Falcon populations in California	(1) Check the California Peregrine Falcon population exhibited source-sink behaviour. (2) Test whether pesticides probably cause for population declines by reducing adult survival in addition to their	Stage-structured matrix model - Population viability analysis (PVA) model	- PVA has been an important tool for evaluating species extinction risk and alternative management strategies. Here tested previously published predictions from a stage-structured PVA of bird species. - Best supported model selected using likelihood index. - Compared trajectories of the bird population to stochastic model	California, USA	- Peregrine Falcons (<i>Falco peregrinus</i>) <i>Conservation status: Least Concern (Population stable)</i> - Listed as endangered in 1970 under the federal Endangered Species Act	Human influence (extensive use of organochlorine pesticides, including DDT)	- Citizen science data sets (Christmas Bird Count, CBC / North American Breeding Bird Survey, BBS) - Systematic census done by the SCPBRG	2006

		long-appreciated effects on reproductive success. (3) Check whether the population would continue to grow with the termination of the captive breeding program because excess birds produced in the healthier northern subpopulation.		simulations to assess agreement of predictions with data.					
35	Full annual cycle climate change vulnerability assessment for migratory birds	Present a CCVA that incorporates climate and life-history data from the full annual cycle and uses migratory connectivity where possible.	Count-based population viability analysis	- CCVAs used to help inform management and conservation decisions such as selecting species or habitats to target for conservation, prioritizing land acquisition, directing monitoring efforts, and identifying specific factors that may contribute to vulnerability. - Used multistate mark-re-encounter models only for two bird species to estimate geographic linkages which estimate the probability that populations from a specific region spend the winter in specific non-breeding regions. - To estimate risk of quasi-extinction, used a count-based population viability analysis developed to estimate extinction risk of rare species of concern.	The Upper Midwest and Great Lakes (UMGL) region, USA	- 46 species of migratory birds breeding in the Upper Midwest and Great Lakes (UMGL) region (Table 1) - 10 species as highly vulnerable to climate change and two as having low vulnerability and the remaining 34 species ranked as moderately vulnerable	- Climate Change - Habitat loss	Two workshops (at the Smithsonian Conservation Biology Institute, Washington, D.C., USA)	October 2012 and January 2013

Article No.	Article Name	Objectives	Statistical model	Analysis description	Cities of study undertaken	Species included	Risk factors	Data sources	Duration
36	Continental-scale decreases in shorebird populations in Australia	Analyse geographical variation in trends, associating them with threats and protective measures operating at shorebird sites to identify elements related to population declines.	Linear multilevel or hierarchical mixed effects model	- Use of Multilevel or hierarchical linear regression has several advantages for analysing patchy datasets, and in this case: (1) it allows direct modelling of the variation among bird species areas; (2) it allows the inclusion of predictors at the bird species area level; (3) it accounts for the spatial hierarchy in the data, which are collected at the count-area resolution grouped by bird species area, and then grouped for all of Australia; (4) it accounts for data that varies in length of time-series and amount of missing data; and (5) it inherently gives more weight to those time-series with larger abundances and less variation.	East Asian–Australasian Flyway (EAAF)	Shorebird populations in wetlands throughout Australia (Table 1)	- Habitat loss - Human influence (human disturbance or agitation of birds and human impacts from aquaculture, management or industrial activity on the environment, diminishing water quality and reduction of available roosting sites) - Land use change (inappropriate water levels for non-tidal wetlands)	Shorebird abundance data from BirdLife Australia's Shorebirds 2020 database	1996-2014
37	Geographic range size and extinction risk assessment in nomadic species	(1) Determined temporal variability in the geographic range size (i.e., AOO) and therefore extinction risk for a suite of Australian nomadic birds. (2) Compared time-sliced estimates of AOO (i.e., monthly estimates based on modelled distribution maps) against more traditional estimates of AOO based on occurrences of taxa pooled across time. (3) Used the results of these models to provide guidelines for incorporating range size variability into existing extinction risk assessments.	Species Distribution Models	- Used the software Maxent to predict the distribution of each species from the occurrence data sets. - Used linear models to analyse the relationship between the pooled geographic range size and the response variables of minimum, maximum, and mean range sizes estimated from our models. - Reclassified the Maxent logistic probability into predictions of absence and probability of presence using equal sensitivity and specificity threshold values. - The area under the receiver operating characteristic curve (AUC) used to evaluate model performance.	Australia Aric-zone	43 arid-zone bird species described as nomadic or possibly nomadic (Table 1)	- Climate change - Habitat loss	Bird occurrences were collated from 20-minute area searches of 2 ha plots	June 2000 to March 2011

*MaxEnt is a machine learning statistical tool that compares well with or outcompetes other modelling technique. It has been shown to be similar to more conventional regression-based approaches used for modelling species environmental correlates and can be applied to a variety of ecological questions depending on how models are calibrated and evaluated.